MAILTRACE AI // FORENSIC TELEMETRY PLATFORM

Digital Evidence & Forensic Incident Dossier

REPORT REFERENCE:	RPT-■20260917-■B2A2C0	**GENERATED (UTC):**	2026-■09-■17 16:■53:■20 UTC
EVIDENCE IDENTIFIER:	sample-■001	**ASSIGNED CASE:**	UNASSIGNED /■ DIRECT EVIDENCE
INVESTIGATOR:	SOC Lead Analyst	**CLASSIFICATION:**	**CONFIDENTIAL // EVIDENTIARY DISCLOSURE**
EMAIL SHA-256:	`97d4b2e811c7520e5e79603f9050■d268159b360b9432df03d4083d8e■57ef228a`	**HASH ALGORITHM:**	FIPS PUB 180-4 SHA-256 (Preserved Literal)

1. Executive Forensic Summary

DETERMINISTIC THREAT SCORE	ASSESSED SEVERITY	ML PHISHING PROBABILITY	ML CLASSIFICATION
82/100	**CRITICAL**	**89.0%**	**PHISHING**

Email risk assessed as Critical (82/100) driven by 8 forensic signal(s): Brand impersonation (+18), Credential harvesting (+12), DMARC failure (+12), Link mismatch (+15), NLP phishing classifier (+10).

2. Email Headers & Sender Alignment

From:	`Microsoft Security &■lt;■security@■micros0ft-■example.■com&■gt;■`
To:	`employee@■company.■com`
Subject:	**URGENT:■ Verify your Microsoft Account**
Date:	Sun,■ 7 Sep 2026 10:■42:■00 +■0530
Reply-To:	`support@■example.■net`
Return-Path:	`bounce@■example.■net`
Message-ID:	`&■lt;■abc123@■example.■com&■gt;■`

From Domain:	`micros0ft-■example.■com`	**Reply-To Domain:**	**example.■net (MISMATCH - SUSPICIOUS)**
Return-Path Domain:	**example.■net (MISMATCH)**	**Alignment Verdict:**	**REPLY-TO / SENDER SPOOFING DETECTED**

3. Email Authentication Findings (SPF / DKIM / DMARC)

PROTOCOL	OBSERVED VERDICT	TECHNICAL DETAILS & EVIDENCE
SPF (Sender Policy)	**FAIL**	spf=■fail (domain of bounce@■example.■net does not designate 203.■0.■113.■25)
DKIM (Cryptographic)	**FAIL**	dkim=■fail header.■i=■@■company.■com
DMARC (Domain Policy)	**FAIL**	dmarc=■fail (p=■reject dis=■none)

Notice: Observed authentication result from supplied headers (unverified by local mail server)

4. Deterministic Threat Score Breakdown

CATEGORY	IMPACT	FORENSIC FINDING & JUSTIFICATION

General	**+18 pts**	N/A
General	**+15 pts**	N/A
General	**+12 pts**	N/A
General	**+12 pts**	N/A
General	**+10 pts**	N/A
General	**+10 pts**	N/A
General	**+8 pts**	N/A
General	**+8 pts**	N/A

5. NLP Content Phishing Classification

Model Name:	TF-■IDF +■ Logistic Regression	**Classification:**	**PHISHING**
Phishing Probability:	**89.0%**	**Model Confidence:**	high
Top Text Features:	`httpurl,■ verify,■ password,■ update,■ account`	**Signal Role:**	Auxiliary ML signal (Non-overriding)

6. AI Analyst Assistant Synthesis

Likely Attack Type:	**Credential Harvesting /■ Brand Impersonation Spear-■Phishing**	**Likely Objective:**	Exfiltrate corporate Microsoft c■redentials via disguised credent■ial-■collection landing pages.■

Executive Synthesis: Suspicious message exhibiting critical credential-harvesting indicators with severe sender spoofing, including failed DMARC and SPF checks alongside a deceptive lookalike domain.

Key Forensic Evidence Identified by LLM:
- Failed DMARC (p=reject) and SPF authentication from sending IP 203.0.113.25
- Lookalike domain micros0ft-example.com imitating Microsoft (registered 18 days ago)
- HTML display anchor showing microsoft.com while pointing to evil-tracker.example
- NLP Phishing Classifier flagged high confidence (89% probability) for credential harvesting language
- Suspicious embedded shortened link (bit.ly/3sample-invoice)

7. Extracted Indicators of Compromise (IOCs)

Observed URLs & Suspicion Scoring:

EXTRACTED URL / TARGET HOST	SCORE	FORENSIC OBSERVATIONS & FLAGS
`https:■/■/■micros0ft-■example.■com/■login`	**25/100**	Potential brand impersonation:■ similar to microsoft.■com (91%■ match),■ Suspicious security/■authentication keywords found:■ login
`http:■/■/■evil-■tracker.■example/■login`	**40/100**	HTML display link mismatch:■ visible text claims &■#■x27;■microsoft.■com&■#■x27;■ but destination links to &■#■x27;■evil-■tracker.■example&■#■x27;■,■ Suspicious security/■authentication keywords found:■ login
`https:■/■/■bit.■ly/■3sample-■invoice`	**20/100**	URL shortener domain detected (b■it.■ly),■ Suspicious security/■authentication keywords found:■ invoice

Observed Attachments & Cryptographic Checksums:

FILENAME	MIME TYPE	SIZE	SHA-256 CHECKSUM
invoice_■2026.■pdf.■exe	application/■octet-■stream	145.0 KB	9f86d081884c7d659a2feaa0■c55ad015a3bf4f1b2b0b822c■d15d6c15b0f00a08
terms_■and_■conditions.■pdf	application/■pdf	82.2 KB	4a5e1e4baab89f3a32518a88■c31bc87f618f76673e2cc77a■b2127b7afdeda33b

8. Network Transmission Path & Topology

Earliest Observable Sending IP:	**198.■51.■100.■12**	**Reverse DNS / Host:**	smtp.■example.■org

Observed Infrastructure Route (Chronological Transmission Sequence):

HOP	FROM (HOST / IP)	BY (RECEIVING RELAY)	TIMESTAMP
#1	smtp.■example.■org (198.■51.■100.■12)	mail.■example.■net (no IP)	Sun,■ 7 Sep 2026 10:■41:■59 +■0530
#2	mail.■example.■net (203.■0.■113.■25)	mx.■company.■com (no IP)	Sun,■ 7 Sep 2026 10:■42:■01 +■0530

Notice: Labeled strictly as observed infrastructure route. Does not represent physical attacker travel path.

9. Infrastructure & Domain Intelligence

IP ADDRESS	LOCATION (CITY/COUNTRY)	ASN / NETWORK	ORGANIZATION / ISP
203.■0.■113.■25	New York,■ United States	AS64512	Observed Hosting Network
198.■51.■100.■12	Frankfurt,■ Germany	AS64512	Observed Hosting Network
198.■51.■100.■44	Amsterdam,■ Netherlands	AS12345	Example Cloud Services
203.■0.■113.■88	San Jose,■ United States	AS64512	Observed Hosting Network

10. Brand Impersonation & Lookalike Domains

SUSPECTED BRAND	SPOOFED DOMAIN	SIMILARITY	TECHNIQUES & ATTACK PROFILE
Microsoft	micros0ft-■example.■com	**91.0%**	character_■substitution,■ brand_■keyword

11. Prescriptive Recommendations & Containment Actions

[ ] 1. Block sending IP 203.0.113.25 and domain micros0ft-example.com on perimeter email gateway
[ ] 2. Revoke active sessions and force password resets for any recipient who accessed the link
[ ] 3. Search SIEM proxy/DNS telemetry for outbound HTTP requests to evil-tracker.example and bit.ly/3sample-invoice
[ ] 4. Purge matching message IDs from all internal Exchange/M365 mailboxes

12. Evidentiary Limitations & Chain-of-Custody Certification

Forensic Chain-of-Custody Certification:
This document is generated by MailTrace AI based on literal RFC-822 evidence headers. Public IP geolocation represents coarse regional estimates based on RIR allocations and does not establish street-level physical locations or the personal identity of the sender. All cryptographic hash sums (SHA-256) were computed over unaltered message byte streams and preserve complete digital non-repudiation chain of custody.

EVIDENTIARY DIGITAL SEAL:	SHA256:97d4b2e811c7520e5e79603f9050■d268159b360b9432df03d4083d8e■57ef228a	**VERIFICATION STATUS:**	**CRYPTOGRAPHICALLY VERIFIED**